

DETECTION AND ALERTING OF UNAUTHORIZED SSH LOGIN ATTEMPTS USING SPLUNK

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Step 1: Connect ssh in the kali:

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ ssh admin@192.168.1.3
admin@192.168.1.3:~$ s password:
Permission denied, please try again.
admin@192.168.1.3:~$ s password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Feb 12 08:21:26 AM UTC 2026

System load:          0.09
Usage of /:           13.0% of 73.72GB
Memory usage:         7%
Swap usage:           0%
Processes:            100
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.3
IPv6 address for enp0s3: 2401:4900:8fe5:bbbf:a00:27ff:fedb:2204

Expanded Security Maintenance for Applications is not enabled.

65 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Feb 12 08:10:06 2026 from 192.168.1.2
```

```
Last login: Thu Feb 12 08:10:06 2026 from 192.168.1.2
admin@ubuntu:~$ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root
[sudo] password for admin:
Splunk> Like an F-18, bro.

Checking prerequisites ...
  Checking http port [8000]: open
  Checking mgmt port [8089]: open
  Checking appserver port [127.0.0.1:8065]: open
  Checking kvstore port [8191]: open
  Checking configuration... Done
New certs have been generated in '/opt/splunk/etc/auth'.
Checking critical directories ... Done
Checking indexes...
  Validated: _audit _configtracker _dm_summary _dsappevent _dsclient _dspho
nehome _internal _introspection _metrics _metrics_rollup _telemetry _thefishbucket histor
y main summary
  Done
Checking filesystem compatibility... Done
Checking conf files for problems... Done
Checking default conf files for edits...
Validating installed files against hashes from '/opt/splunk/splunk-10.2.0-d749cb1
7ea65-linux-amd64-manifest'
All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
Done

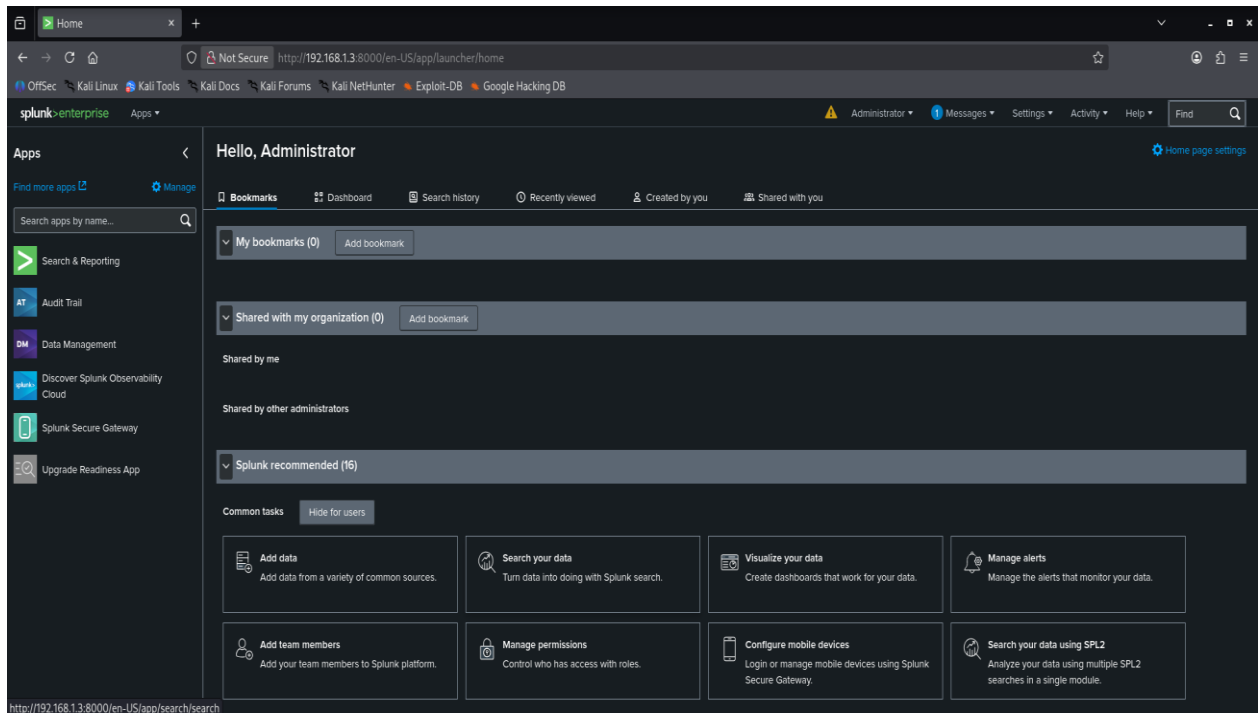
Waiting for web server at http://127.0.0.1:8000 to be available.....
..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://ubuntu:8000
admin@ubuntu:~$ █
```

If you give wrong password here that was fine. it's used for after some time.

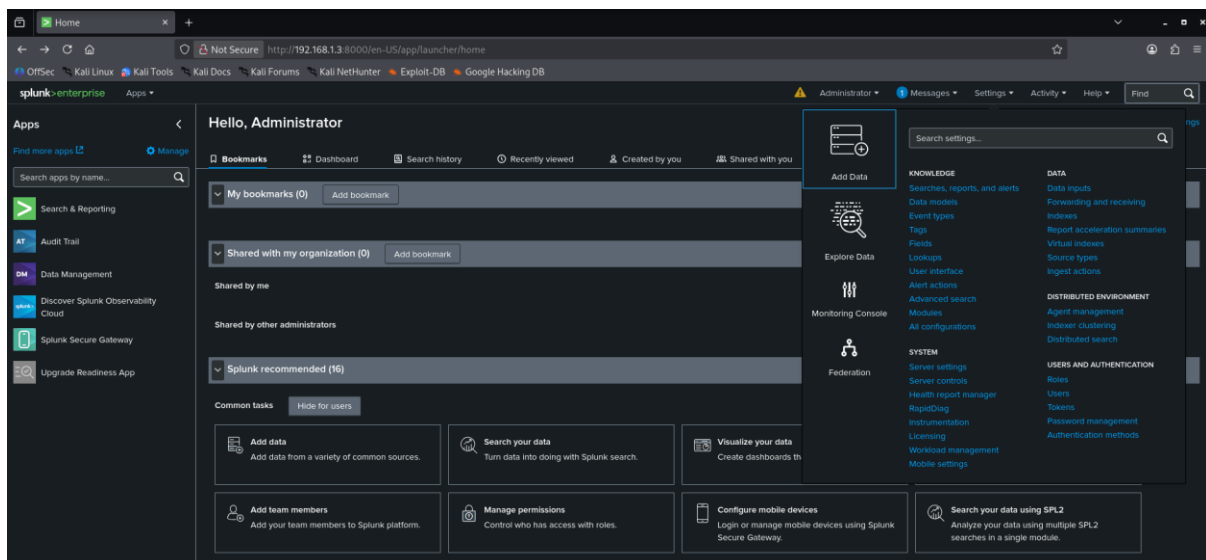
Step 2: login Splunk:



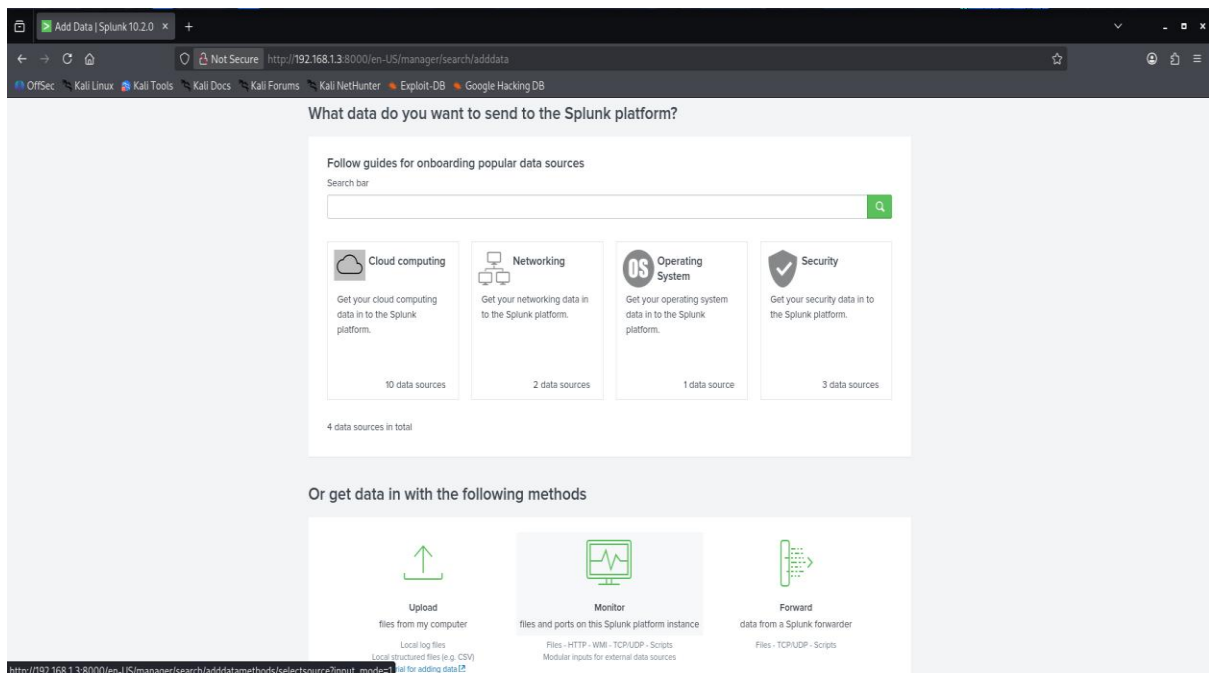
We need to login Splunk. It's opened above mentioned page.

Step 3: Make Sure Ubuntu SSH Logs Are Ingested into Splunk:

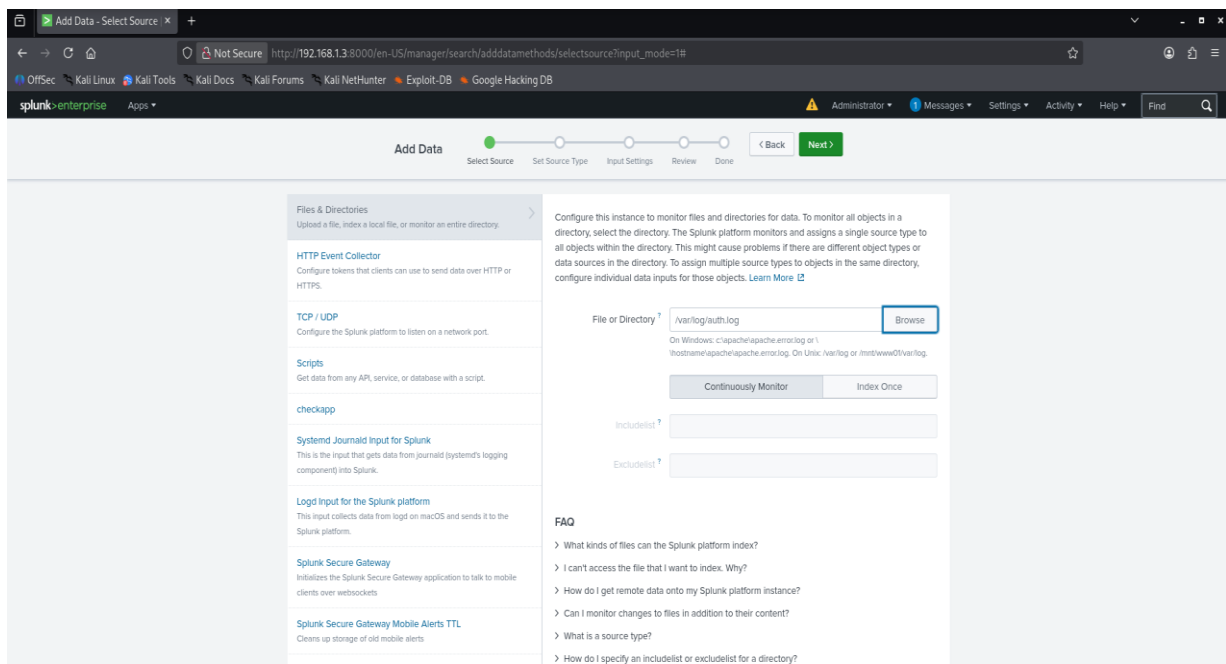
i) Go to settings → Click → Add Data



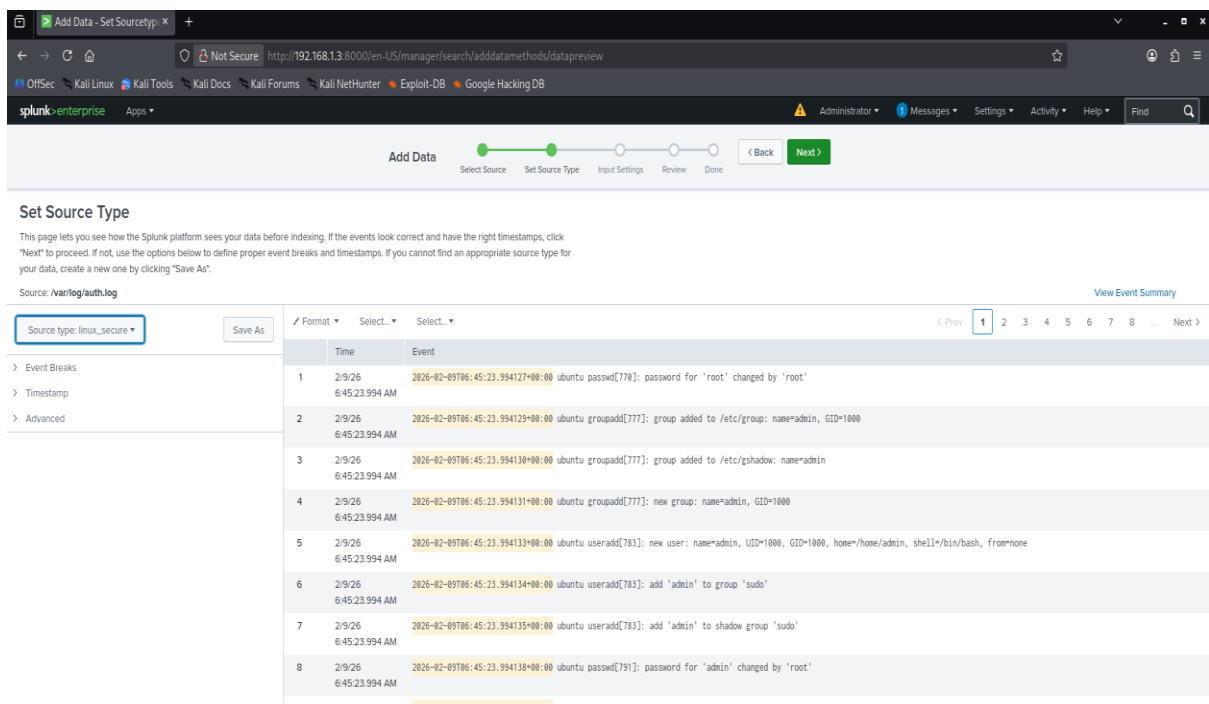
ii) A page will open as shown below



iii) Click → Monitor → Files & Directories → File or Directory: /var/log/auth.log → Next



iv) Change → Source Type : Operating System → Linux Secure → Next



Add Data - Set Sourcetype

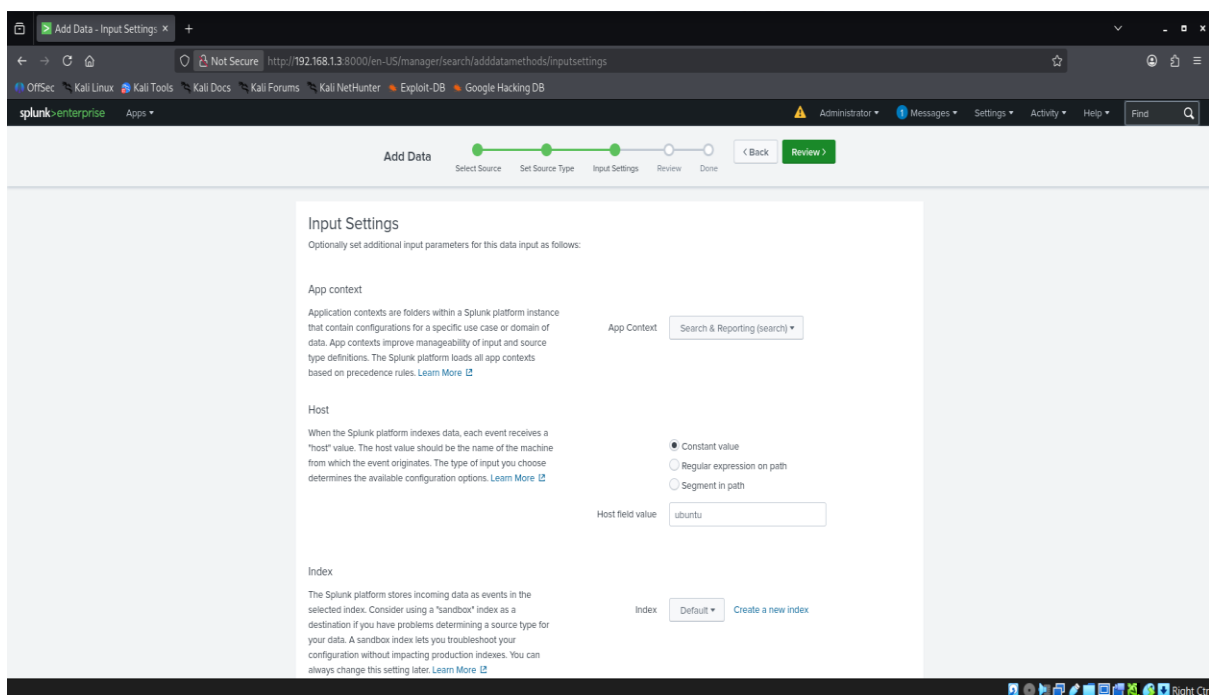
Source: `/var/log/auth.log`

Source type: `linux_secure` [Save As]

View Event Summary

Time	Event
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994127+00:00 ubuntu passwd[770]: password for 'root' changed by 'root'
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994125+00:00 ubuntu groupadd[777]: group added to /etc/group: name=admin, GID=1000
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994130+00:00 ubuntu groupadd[777]: group added to /etc/gshadow: name=admin
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994131+00:00 ubuntu groupadd[777]: new group: name=admin, GID=1000
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994133+00:00 ubuntu useradd[783]: new user: name=admin, UID=1000, GID=1000, home=/home/admin, shell=/bin/bash, from=none
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994134+00:00 ubuntu useradd[783]: add 'admin' to group 'sudo'
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994135+00:00 ubuntu useradd[783]: add 'admin' to shadow group 'sudo'
2/9/26 6:45:23.994 AM	2026-02-09T06:45:23.994138+00:00 ubuntu passwd[791]: password for 'admin' changed by 'root'

v) Click → review



Add Data - Input Settings

Input Settings

Optionally set additional input parameters for this data input as follows:

App context

Application contexts are folders within a Splunk platform instance that contain configurations for a specific use case or domain of data. App contexts improve manageability of input and source type definitions. The Splunk platform loads all app contexts based on precedence rules. [Learn More](#)

App Context: Search & Reporting (search) ▼

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

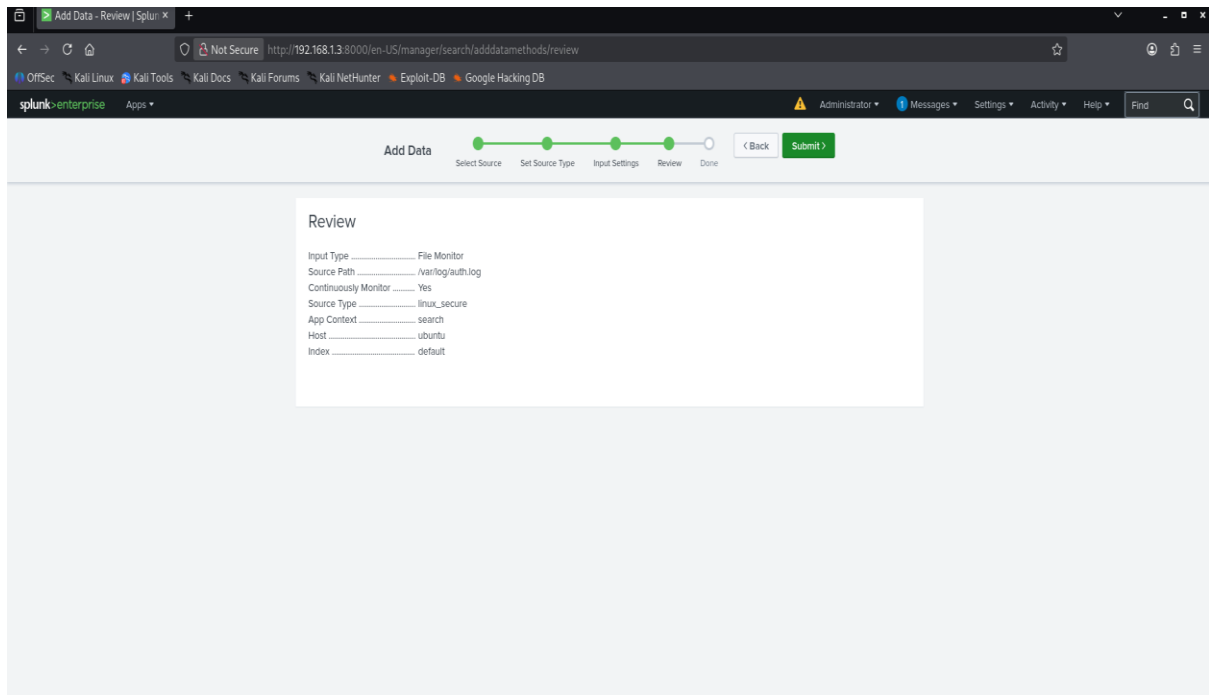
Host field value: ubuntu

Index

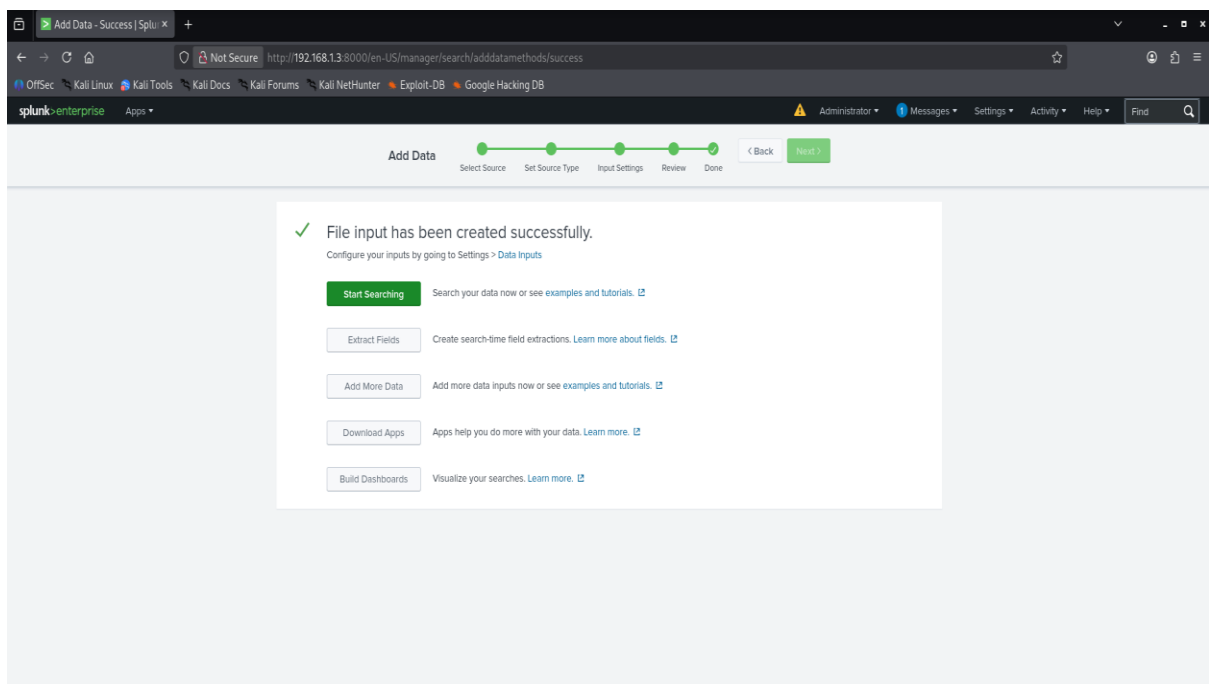
The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index: Default ▼ [Create a new index](#)

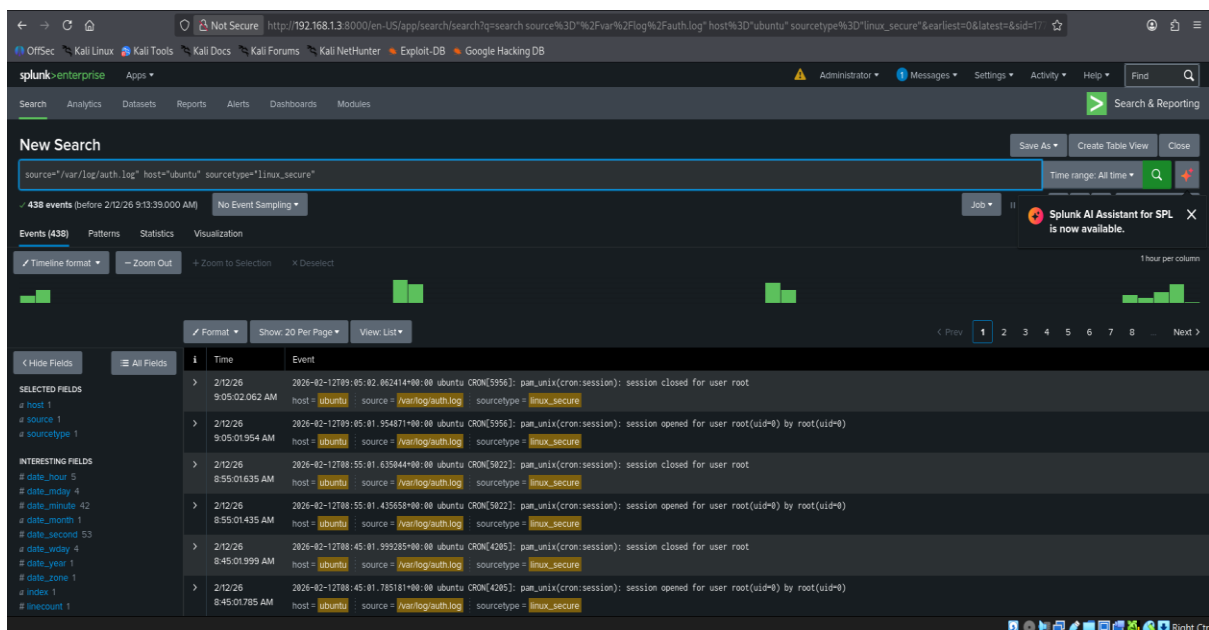
vi) Click → submit



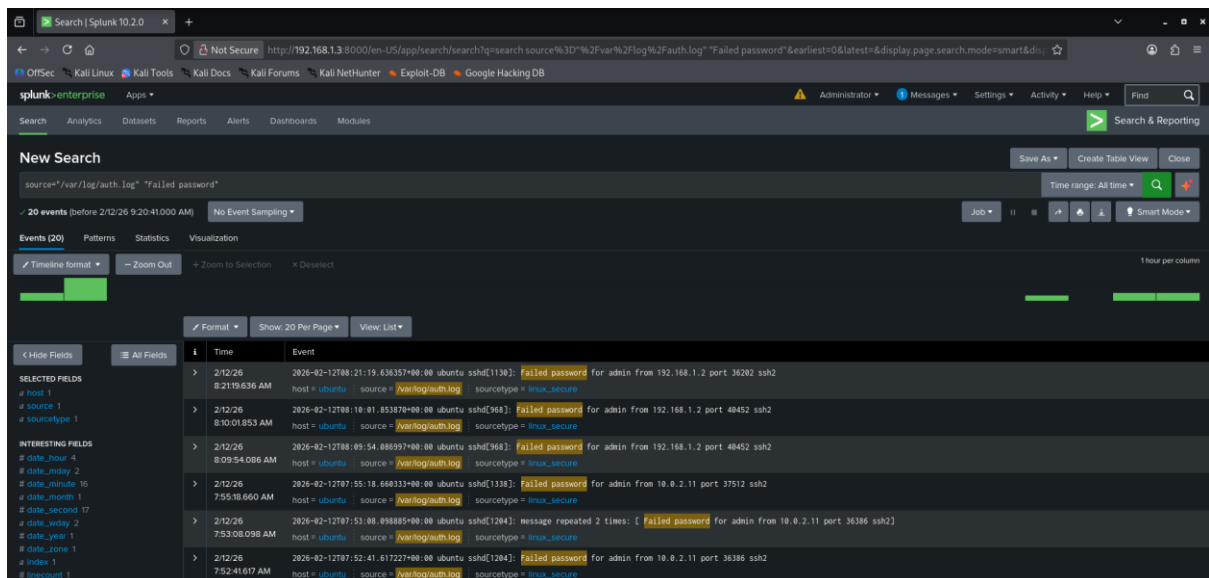
vii) Click → Start searching



i) A page will open as shown below



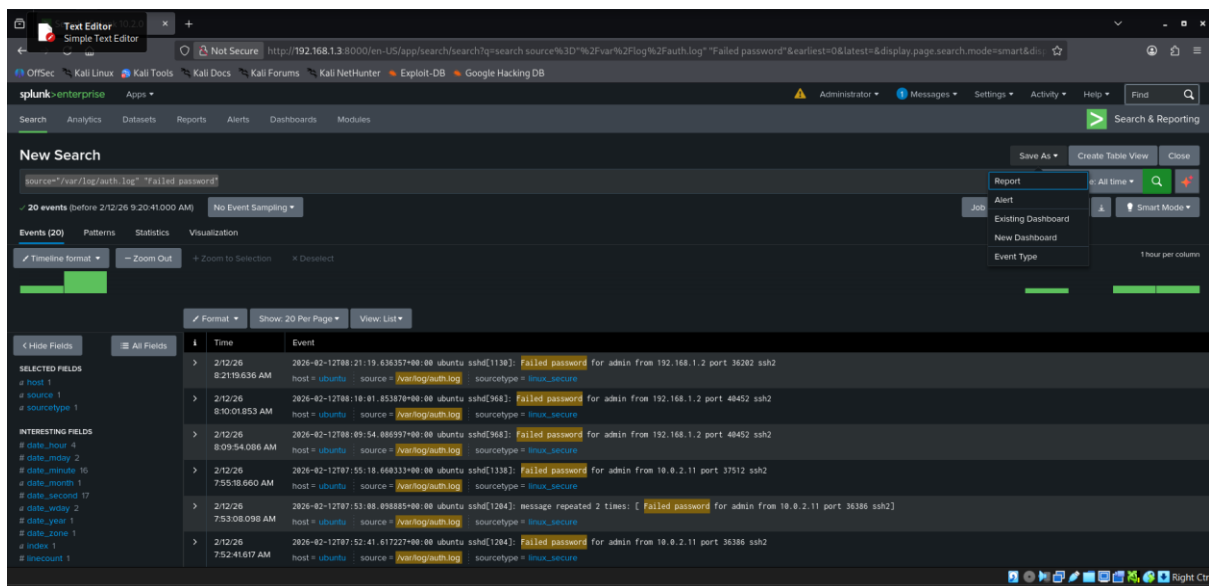
ii) Search for Failed SSH Login Attempts:
Go to New search → Type →
source=\"/var/log/auth.log\" \"Failed password\" →
Search



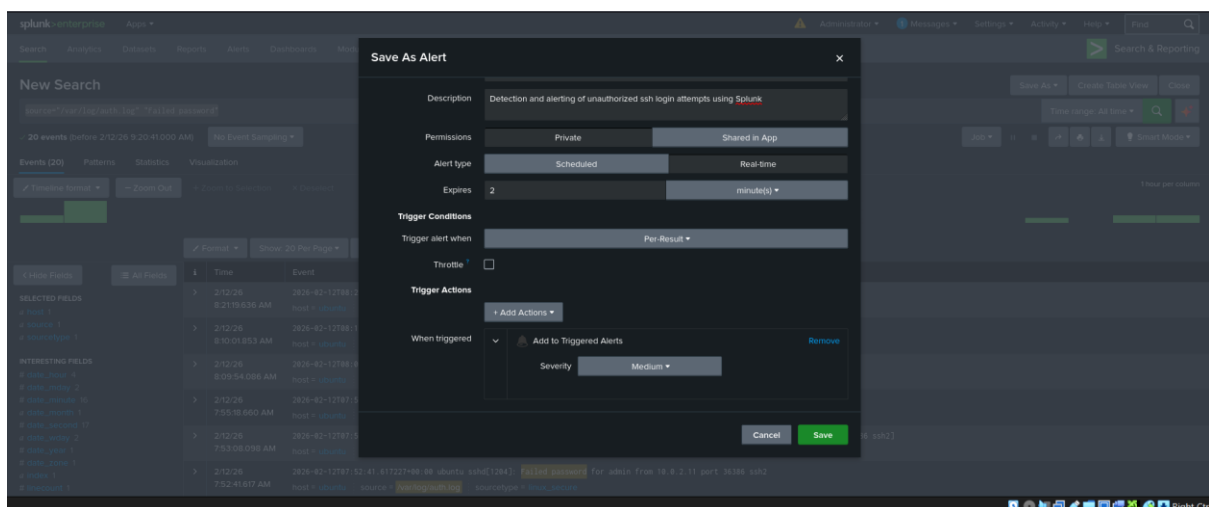
If you search for the term ‘Failed password’ in Splunk, it will display all recorded events where incorrect password attempts were made.

Step 4: Failed log Alert Settings:

i) Click → Save as → Alert



ii) Set Alert Actions



Title : Failed log Attempt

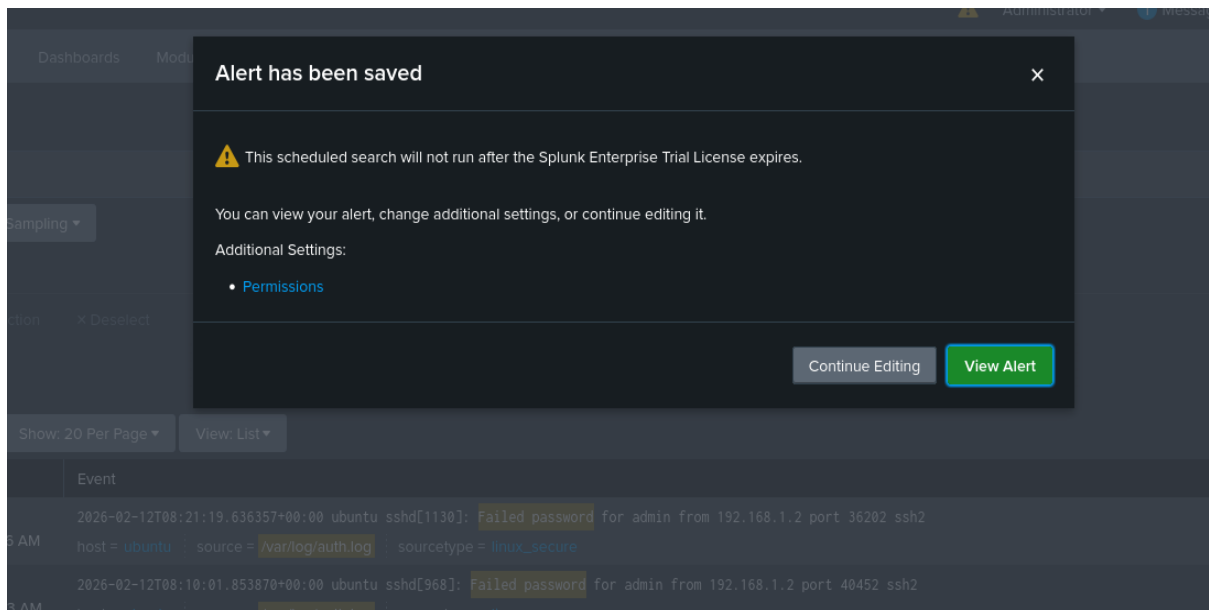
Description : Detection and alerting of unauthorized ssh login attempts using Splunk

Alert type : Real time

Expires : 2 minutes

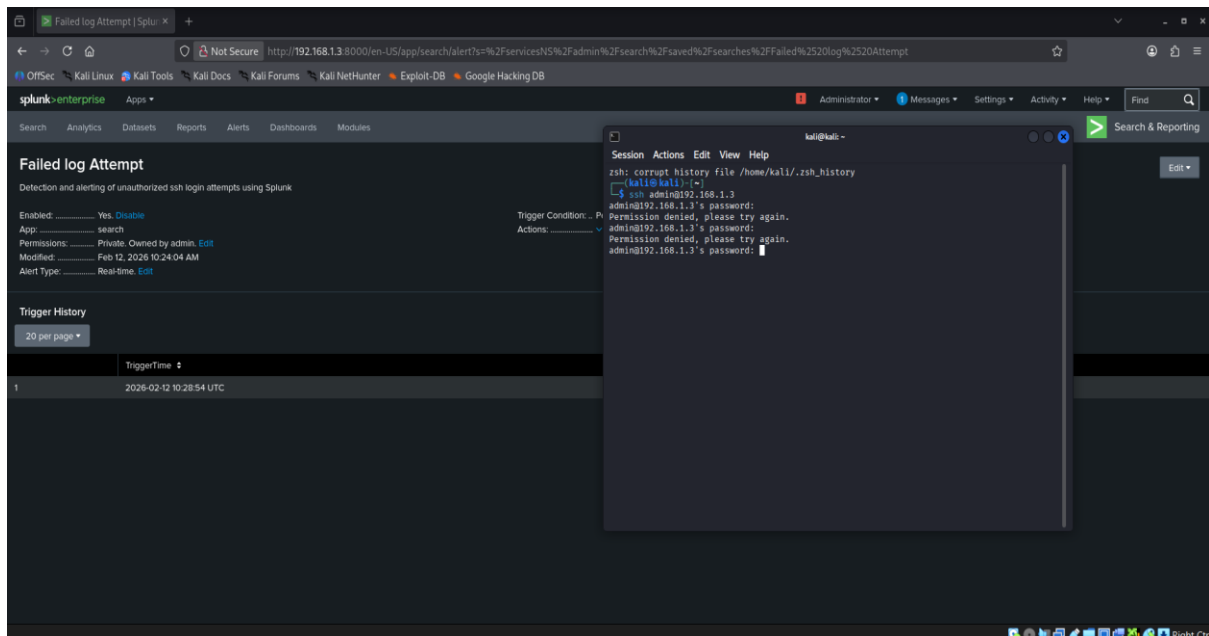
Trigger Actions : Add actions → Select → Add to Triggered Alerts → Severity: Medium → Save

iii) A pop-up show



○ Alert has been saved → View Alert

iv) Failed log Attempt



If you are trying to log in with the wrong password in the ssh, it's shows alert in the Splunk alert page.

v) Track unauthorized SSH access attempts

The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `source="/var/log/auth.log" "failed password"`. The search results show 1 event from 11/70 12:00:00.000 AM to 2/12/26 10:28:54.655 AM. The event details are as follows:

Type	Field	Value	Actions
Selected	host	ubuntu	
	source	/var/log/auth.log	
	sourcetype	linux_secure	
Time	_time	2026-02-12T10:28:52.766-00:00	
Default	index	main	
	splunk_server	ubuntu	

Click → View Results

Conclusion:

By setting up this failed login alert, an immediate notification will be generated whenever an unknown person attempts to log in. This allows us to quickly identify the attacker's details and take prompt action to prevent unauthorized access.